

The Matrix Multiplication Bound

$\omega \leq 9/4$ over Arbitrary Fields

Abstract

OpenAI proved that the matrix multiplication exponent over the complex numbers is at most $9/4$. We extend that argument to every field. The extension replaces a Fourier period that can vanish in positive characteristic, removes an integer-node choice from the Lean interpolation proof, and descends exact tensor decompositions from an algebraic closure with a fixed overhead. We describe these changes and their formalization, taking the underlying $9/4$ construction from OpenAI’s proof.

1 Statement and relation to the original result

For a field F , write $\omega(F)$ for the infimum of the real numbers τ such that, for every $\varepsilon > 0$, square matrices of size n over F can be multiplied by division-free arithmetic circuits of cost $O_{F,\varepsilon}(n^{\tau+\varepsilon})$. Scalar additions, subtractions, and multiplications each cost one; inputs and constant loads are free. Circuits and their constants may depend on the size, and the constant implicit in the bound may depend on F and ε .

Theorem 1.1. *For every field F and every $\varepsilon > 0$, there is a constant $C > 0$ such that, for every integer $n \geq 1$, there is a division-free arithmetic circuit over F computing $n \times n$ matrix multiplication with at most $Cn^{9/4+\varepsilon}$ operations. In particular,*

$$\omega(F) \leq \frac{9}{4}.$$

The bound is obtained from exact identities of tensor coefficients.

The original bound over $\mathbb{C}$, its auxiliary-separation construction, and its spectral and polynomial-growth argument are due to OpenAI [1]. This note concerns the additional field scope. We use the manuscript and Lean sources at the pinned revision in [1, 2]; the formal extension is archived in [3].

The characteristic-zero consequence is already implicit in [1, Remark 5.2]. For a fixed finite rank decomposition, its coefficient equations have rational coefficients. A complex solution therefore gives a solution over $\overline{\mathbb{Q}}$, and the finitely many coefficients lie in a number field. Coordinate simulation over $\mathbb{Q}$ has constant overhead once the decomposition is fixed. The resulting rational coefficient identities extend to every characteristic-zero field. The extension degree may depend on ε , which is permitted in the definition of the exponent.

This reasoning does not by itself cover every positive characteristic: reducing a rational scheme modulo a prime can fail at its denominators, and the exceptional primes can depend on the scheme chosen for ε . Instead, for each field F we work directly over $K = \overline{F}$. Section 2 explains the modifications over K ; Section 3 descends the result to F .

We distinguish the circuit exponent from the exact coefficient-rank exponent. Let

$$T_n = \sum_{i,j,k=1}^n x_{ij}y_{jk}z_{ki}, \quad \nu(F) = \inf_{n \geq 2} \log_n R_F(T_n),$$

where R_F is tensor rank over F . The proof gives $\nu(F) \leq 9/4$ and then $\omega(F) \leq \nu(F)$. For unrestricted arithmetic circuits over finite fields, functional correctness alone need not imply a formal polynomial identity; the tensor argument proves exact coefficient identities.

2 The construction in positive characteristic

Fix an algebraically closed field K . Such a field is infinite, regardless of its characteristic. We retain the tensor restrictions, degenerations, and real-valued tensor characters of [1].

2.1 A nonvanishing Fourier period

The finite separation construction in [1, Proposition 3.1] uses a Fourier period $5M$ for $M \geq 1$ blocks. Normalization divides by this period. In characteristic p this is invalid when $p \mid 5M$; in that case there is also no primitive $(5M)$ -th root of unity. Replace the period by

$$L = L_K(M) = \begin{cases} 5M, & (5M)1_K \neq 0, \\ 5M + 1, & (5M)1_K = 0. \end{cases} \quad (2.1)$$

Then $5M \leq L \leq 6M$ and $L1_K \neq 0$, since consecutive integers cannot both have zero image in a field. The polynomial $X^L - 1$ splits with distinct roots in K . Its roots form a finite subgroup of $K^\times$, hence a cyclic group, so K contains a primitive L -th root ζ .

Lemma 2.1. *For $e \in \mathbb{Z}$,*

$$\frac{1}{L} \sum_{r=0}^{L-1} \zeta^{re} = \begin{cases} 1, & L \mid e, \\ 0, & L \nmid e. \end{cases}$$

If $|e| < L$, the value is 1 precisely when $e = 0$.

Proof. If $L \mid e$, all summands equal 1. Otherwise $\zeta^e \neq 1$, and the geometric-sum identity gives zero. Both divisions are valid: $L1_K$ and $\zeta^e - 1$ are nonzero in their respective cases. The last statement is an assertion about divisibility of integers. $\square$

In the substitutions of [1, Proposition 3.1], the Fourier phase is $e = u - v + 2(g - h)$, with all four labels in $\{1, \dots, M\}$. It satisfies $|e| \leq 3(M - 1) < L$. Thus Lemma 2.1 retains exactly the same terms as the original filter. The unchanged weight calculation assigns total weight $(g - h)^2$ to a surviving term; its zero-weight part forces $g = h$ and $u = v$. These weights are integer exponents of a formal parameter, so they are not reduced modulo the characteristic. In particular the argument also works in characteristic two.

The resulting separation uses L copies of the source tensor instead of $5M$ copies. This changes a bounded finite overhead, not the entropy inequality. Indeed, in the type-counting step of [1, Corollary 3.2], let M_N be the number of words of a fixed rational type q , where N runs through multiples of a common denominator of q , and put $C_N = \prod_i \lambda(A_i)^{Nq_i}$. Here the A_i are the sectors of A , λ is a tensor character, and $\lambda(B_X(m)) = m^{p_X}$ for the oriented dot-product tensor. Separation and degeneration monotonicity give

$$6M_N \lambda(A)^N \geq L_K(M_N) \lambda(A)^N \geq M_N^{1+p_X} C_N.$$

After canceling M_N and taking N -th roots, the extra factor is $6^{1/N} \rightarrow 1$. The same entropy limit and continuity argument therefore give the original inequality. All quantities in this limit are real numbers, not elements of K .

2.2 Interpolation over the algebraic closure

The manuscript's interpolation argument already allows arbitrary distinct nonzero nodes [1, Lemma 2.3]. The corresponding original Lean implementation chooses the images of $1, 2, \dots, D + 1$ in $\mathbb{C}$. That implementation choice must be replaced in positive characteristic: integer images can coincide or vanish. An infinite field supplies any finite number of distinct nonzero nodes, with invertible pairwise differences.

For clarity, the part of interpolation used here can be expressed without characteristic assumptions as follows. Suppose a polynomial tensor family $G(t)$ has degree at most D , has $G(0) = B$, and every $G(t)$ with $t \neq 0$ is a restriction of A . For each $m \geq 1$, the family $G(t)^{\otimes m}$ has degree at most Dm and constant coefficient $B^{\otimes m}$. Interpolating at $Dm + 1$ distinct nonzero nodes expresses that coefficient as a linear combination of the evaluations. Hence

$$\lambda(B)^m \leq (Dm + 1)\lambda(A)^m.$$

Taking m -th roots proves $\lambda(B) \leq \lambda(A)$. If a degeneration initially has a nonzero leading degree, divide out that power of the parameter before applying this argument; the nonzero-node condition permits this normalization. The same choice of nodes gives the usual evaluation and interpolation bound for polynomial multiplication over K .

2.3 The retained 9/4 argument

With these two choices made, the remaining proof of [1] applies over K . The determinant filtration uses polynomial-domain identities and bases with unit pivots; it does not require division by the characteristic or a semisimple splitting. The sector construction uses integer indices and weights. The real growth inequalities in [1, Sections 4–5] have no dependence on the characteristic of the coefficient field. Finally, the separation and fixed-point arguments in [1, Appendix A] act on real-valued functions on the tensor semiring. They do not impose a topology on K . A natural number in that semiring counts direct summands and is not its image in K .

Consequently the detecting-character argument, with the same polynomial-growth estimate, gives

$$\nu(K) \leq \frac{9}{4} \quad (K \text{ algebraically closed}). \quad (2.2)$$

This step uses the original proof rather than a new derivation of the numerical bound. In the formalization, its endpoint is `AuxiliarySeparation.exactRankExponent_le_nine_quarters`.

3 Descent with a fixed coefficient algebra

Field-extension invariance is classical; see Schönhage's rank-descent argument and characteristic-invariance theorem [4, Theorems 2.7–2.8]. The descent principle below isolates the order of quantifiers used in our formalization. It uses finite-dimensional linear algebra and applies without a separability assumption.

Lemma 3.1. *Let S be a nonzero commutative F -algebra of finite dimension d . If a finite coefficient tensor T over F has an r -term rank-one decomposition after scalar extension to S , then*

$$R_F(T) \leq d^2 r.$$

Proof. Choose an F -basis $e_1, \dots, e_d$ of S and an F -linear map $\pi : S \rightarrow F$ with $\pi(1) = 1$. Such a map exists because $1 \neq 0$; extend 1 to a basis and take its coordinate functional. Thus $\pi(c1_S) = c$ for $c \in F$.

Write the decomposition as $T(x, y, z)1_S = \sum_{\ell=1}^r a_\ell(x)b_\ell(y)c_\ell(z)$. Expand the first two factors in the chosen basis: $a_\ell(x) = \sum_i a_{\ell i}(x)e_i$ and $b_\ell(y) = \sum_j b_{\ell j}(y)e_j$, with coefficients in F . Applying π gives the coefficient identity

$$T(x, y, z) = \sum_{\ell=1}^r \sum_{i,j=1}^d a_{\ell i}(x)b_{\ell j}(y)\pi(e_i e_j c_\ell(z)).$$

Each summand separates the three coordinates and has coefficients in F . There are at most rd^2 summands. No multiplicativity of π or field trace is used. $\square$

Proposition 3.2. *Let E/F be an algebraic field extension, and let T be a finite coefficient tensor over F . If $R_E(T_E) \leq r$, then there exists an integer $d \geq 1$ such that, for every $m \geq 1$,*

$$R_F(T^{\otimes m}) \leq d^2 r^m. \quad (3.1)$$

Proof. Fix one r -term decomposition over E . Its finitely many coefficients generate an F -subalgebra $S \subseteq E$ of finite dimension d , because each generator is algebraic over F . The chosen decomposition already exists over S , and taking its m -th tensor power uses only products of coefficients in the same algebra S . It therefore has at most r^m terms over S . Apply Lemma 3.1 to this powered decomposition. $\square$

The order is $\exists d \forall m$: the coefficient algebra is fixed before the tensor power varies. Descending one block first and then powering its descended decomposition would instead give the weaker bound $(d^2 r)^m$. Equation (3.1) avoids charging the extension cost at each recursive level. The proof works for inseparable extensions and for imperfect base fields.

Corollary 3.3. *For every algebraic field extension E/F , $\nu(F) \leq \nu(E)$. In particular $\nu(F) \leq \nu(\overline{F})$.*

Proof. Fix $u \geq 2$ and a decomposition of T_u over E of length $r = R_E(T_u)$. Since $T_u^{\otimes m}$ is T_{u^m} after reindexing, Proposition 3.2 yields, for every $m \geq 1$,

$$\nu(F) \leq \log_{u^m} R_F(T_{u^m}) \leq \log_u r + \frac{2 \log d}{m \log u}.$$

Let $m \rightarrow \infty$, then take the infimum over $u \geq 2$. This does not require the infimum defining $\nu(E)$ to be attained. $\square$

4 Arithmetic circuits and the formal statement

Proof of Theorem 1.1. Apply (2.2) to $K = \overline{F}$ and then Corollary 3.3 to obtain $\nu(F) \leq 9/4$. Fix $0 < \delta < \varepsilon$. The definition of the infimum supplies a block size $u \geq 2$ and an exact rank decomposition over F of length $r < u^{9/4+\delta}$.

The generic block-program construction from the original Lean development [2] implements this bilinear formula recursively, including the scalar operations in its linear combinations. On powers of u , its cost satisfies

$$S(uv) \leq rS(v) + 6ru^2v^2, \quad S(1) = 1.$$

The flattening bound gives $r \geq u^2$. Thus the cost is $O(N^{\log_u r})$ when $r > u^2$, and $O(N^2 \log N)$ when $r = u^2$. Padding to a power of the fixed u and using the positive exponent slack gives $O_{F,\varepsilon}(n^{9/4+\varepsilon})$ for every $n \geq 1$. $\square$

The archived formalization [3], written in Lean 4 [5] using Mathlib [6], exports the theorem in namespace `OAI.MatrixMultiplication`. With the real type written as `Real`, its statement is

```
theorem omega_le_nine_quarters (F : Type*) [Field F] :
  Arithmetic.omega F <= (9 : Real) / 4
```

The original statement substitutes $\mathbb{C}$ for F . The definitions of the arithmetic model are preserved; the current source includes only a mechanical module-system port of the protected upstream specification. The original complex conclusion remains a specialization.

Two supporting results make the scope explicit. The exact-rank theorem proves $\nu(F) \leq 9/4$ using equality at every tensor coordinate. The arithmetic theorem has quantifiers

$$\forall \varepsilon > 0 \exists C > 0 \forall n \geq 1 \exists P : \quad P \text{ correctly multiplies over } F, \quad \text{cost}(P) \leq Cn^{9/4+\varepsilon}.$$

Here P must work for every input pair of its size. Over finite fields, its correctness predicate is functional; the separate exact coefficient theorem supplies the stronger algebraic guarantee. The formal result is a family of arithmetic circuits. It does not assert an $O(n^{9/4})$ endpoint bound, effective uniform generation, or a bit-complexity bound.

The source correspondence is short. Within the matrix-multiplication subtree of [3], the Fourier changes are in `AuxiliarySeparation/Separation`; interpolation is in `AuxiliarySeparation/Polynomial/Interpolation.lean`. The two descent results are implemented in `Arithmetic/FieldDescent.lean`. The exponent comparison is in `AuxiliarySeparation/Arithmetic/FieldExtension.lean`. The public entry point is `AllFields.lean`.

The versioned archive records the proof snapshot and its mechanical verification. Its six Comparator claims include the arithmetic bound, infimum side conditions, and explicit exact coefficient identities. The recorded Palomar run checks those claims using Lean and the independent kernels listed in its report. The allowed axioms are `propext`, `Classical.choice`, and `Quot.sound`. These are verification claims about the recorded source snapshot, not a claim of human peer review or a check of this prose document.

Attribution and development. OpenAI supplied the original $9/4$ construction and complex-field proof [1, 2]. The contribution described here is its adaptation to arbitrary positive characteristic, the explicit descent argument, and their formal verification. No historical-priority claim is made for the scalar-descent principle. The extension and Lean formalization were developed with consumer-accessible GPT-6 Astra and GPT-6.1 Sol through Codex, under human direction. The repository records that provenance. This note was prepared with AI assistance.

References

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